LETTERS
PUBLISHED ONLINE: 12 AUGUST 2012 | DOI: 10.1038/NPHOTON.2012.178
nature photonics

Three-dimensional coherent X-ray surface scattering imaging near total external reflection

Tao Sun[1]*, Zhang Jiang[1], Joseph Strzalka[1], Leonidas Ocola[2] and Jin Wang[1]*

Lensless X-ray coherent diffraction imaging (CDI) has emerged as a thriving field promising applications in materials and biological sciences[1–13] with a theoretical imaging resolution only limited by the X-ray wavelength. Most CDI methods use transmission geometry, which is not suitable for nanostructures grown on opaque substrates or for objects of interest comprising only surfaces or interfaces. Attempts have been made to perform CDI experiments in reflection geometry, both optically and with X-rays, but the reconstruction resulted in mostly planar images, with less success in the third dimension[14,15]. Here, we discuss the development of coherent surface scattering imaging in grazing-incidence geometry that takes advantage of enhanced X-ray surface scattering and interference near total external reflection. We demonstrate the successful reconstruction of substrate-supported non-periodic surface patterns in three dimensions with nanometre resolution in the direction normal to the substrate, promising wide applications in elucidating structures in substrate-supported and buried nanoelectronics and photonics.

For samples such as thin films, nanopatterns and buried interfaces, reflection-mode coherent diffraction imaging (CDI) can be more successful than transmission-mode CDI. Previous studies using reflection-mode CDI implied that the results were not sufficient to generate true three-dimensional reconstruction of nanostructures at surfaces. To obtain unambiguous structural information in the third dimension normal to the substrate, we collected coherent surface scattering patterns from substrate-supported nanostructures at grazing-incidence angles in the vicinity of the sample critical angle (typically a fraction of a degree for hard X-rays), as shown in Fig. 1a. In this angular region, in-plane scattering is greatly enhanced as a result of total external reflection. As demonstrated in many X-ray studies, including reflectivity and diffuse scattering at grazing incidence, surface scattering is particularly sensitive to the sample electron density distribution in the plane and in the direction perpendicular to the substrate surface[16–20], which allows depth-resolved structural characterization to be achieved by varying the incident angle. At grazing incidence, the momentum transfer along the surface normal (z or q_z direction) spans several orders of magnitude from 0.001 to $>1\ \mathrm{nm}^{-1}$, which enables high-resolution reconstruction of sample structures along the surface normal direction. At angles far beyond the critical angle, the scattering intensity (either coherent or incoherent) decreases algebraically with incident angle (or q_z). The grazing-incidence coherent surface scattering imaging (CSSI) technique is therefore particularly suitable for weakly scattering non-periodic objects, as it ensures a large signal-to-noise ratio within a reasonable data collection time. Another advantage of grazing-incidence CSSI over high-angle reflection-mode CDI is its larger detectable area (a consequence of geometry), albeit with a correspondingly reduced resolution. All these unique scattering characteristics in the near total external reflection region promise an unprecedented surface sensitivity for mapping scattering intensity in reciprocal space and reconstructing the corresponding surface structures in three dimensions.

To demonstrate the grazing-incidence CSSI concept, we first examined a simple gold double-stripe structure, patterned on a 3-mm-thick flat silicon substrate along the x-direction. Figure 1b shows the scattering pattern obtained at an incident angle of $\alpha_i = 0.6^\circ$, which is above the silicon critical angle (0.245° at $E = 7.35$ keV) and just below the gold critical angle (0.61°). In contrast to conventional transmission CDI, where most incident X-rays penetrate through the sample without participating in diffraction events, the specular reflection (at q_x and $q_y \approx 0$) in CSSI is part of the sample scattering pattern, similar to the results in Bragg-diffraction-based CDI[4]. Accordingly, no beam stop is required to block the spot, which allows low-q information to be limited only by the instrument resolution ($\delta q_x \approx 7.6 \times 10^{-6}\ \mathrm{nm}^{-1}$, $\delta q_y \approx 7.6 \times 10^{-4}\ \mathrm{nm}^{-1}$). The double-stripe configuration is analogous to that of the classic Young's double-slit experiment, where waves from two slits interfere to generate intensity modulations on the detection screen; however, there are several major different and also distinct characteristics. First, in addition to the X-rays being strongly reflected by the two gold stripes, weak scattering by the silicon substrate contributes to the interference process. Second, with the grazing-incidence geometry, the horizontal axis of the scattering pattern represents the in-plane y-component of momentum transfer in the small-angle region, and the vertical axis corresponds to both q_z and q_x components. Also, unlike in the transmission geometry, the q_x component is not linearly represented by the vertical pixel position because of the way the Ewald sphere intercepts the reciprocal lattice (for a detailed explanation see Supplementary Section S1). This causes the scattering pattern shown in Fig. 1b to bend downwards at the large azimuthal scattering angle $2\theta_f$.

Figure 1c (open circles) shows the scattering intensity profile along q_y, a line scan at $q_x = 0$ from the scattering pattern displayed in Fig. 1b. The simulation was based on Young's double-slit experiment with a partially coherent X-ray beam. Numerically, the partial coherence was taken into account by the convolution of the instrument resolution and the ideal scattering curve[21]. Details of the simulation are presented in Supplementary Section S2. The remarkably good agreement between the data and the simulation demonstrates the feasibility of coherent surface scattering in the grazing-incidence geometry. One can see that the projected transverse coherence length extends over the millimetre-long stripes along the beam direction. Similar to conventional CDI methods, the CSSI scattering pattern can also be used for reconstructing the exit surface wave (ESW) by using phase-retrieval algorithms, such as hybrid input–output (HIO)[22] and error reduction (ER)[23]. For reconstruction of a planar image on the x–y plane, the data must first be treated to convert the original scattering pattern into one

[1]X-ray Science Division, Advanced Photon Source, Argonne National Laboratory, Argonne, Illinois 60439, USA, [2]Center for Nanoscale Materials, Argonne National Laboratory, Argonne, Illinois 60439, USA. *e-mail: taosun@aps.anl.gov; wangj@aps.anl.gov

© 2012 Macmillan Publishers Limited. All rights reserved.

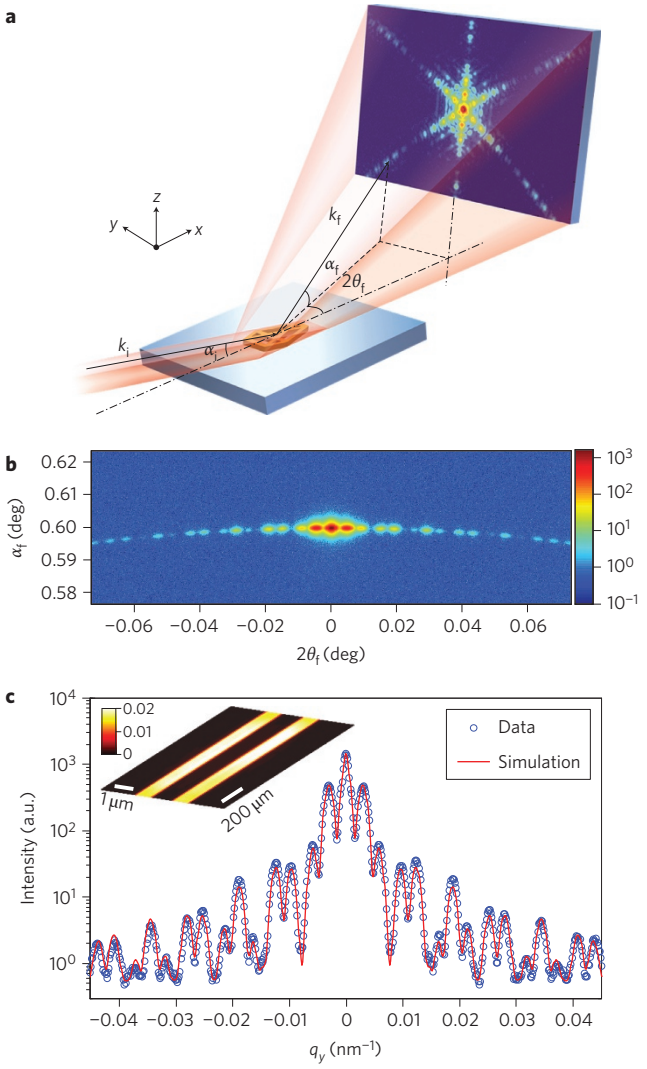


Figure 1 | X-ray coherent surface scattering in grazing-incidence geometry. **a**, Schematic of the experiment with coordinate systems and incident (α_i), reflected (α_f) and azimuthal ($2\theta_f$) scattering angles. **b**, Scattering pattern (logarithmic scale; units = counts) from a gold double-stripe structure. **c**, Intensity profile along the q_y direction (open circles) from the coherent scattering pattern in **b** at $q_x = 0$, and corresponding simulation (solid line). Inset: reconstructed planar ESW amplitude with intensity scale.

with orthogonal q_x–q_y coordinates (shown in Supplementary Section S2). The reconstructed image (that is, the amplitude of the ESW) of the double-stripe gold pattern is shown in the inset of Fig. 1c. Note that the pixel resolutions along the horizontal and vertical directions are different because of the grazing-incident angle.

Having demonstrated the applicability of CSSI, we now show that it can be used to reconstruct more complex non-periodic surface structures. Given a large projected transverse coherence length along the x-direction, we prepared sample patterns with extended features spanning hundreds of micrometres along that direction. Figure 2 presents a scanning electron microscopy (SEM) image (Fig. 2a) and a schematic (Fig. 2b) of a microsized gold outline pattern of the logo of the authors' institute, Argonne National Laboratory. Figure 2c shows the scattering data obtained from the logo structure at an incident angle of $\alpha_i = 0.55^\circ$. The pattern was stitched together with frames from nine different detector positions to extend the reciprocal-space data range. Surface patterns of the 'A', 'N' and 'L' letter structures were also fabricated, and the SEM images and CSSI data are shown in Supplementary Section S4.

With these complex patterns, scattering speckles cover relatively large ranges of q_x and q_z, both of which are represented in the vertical directions in the scattering patterns. In addition, even at constant q_x, the scattering intensity drops as the incident or exit angle increases. Accordingly, the isolation of q_x and q_z contributions and the performance of intensity normalization with respect to the incident and exit angles are prerequisites only for the two-dimensional image reconstruction (see below and Supplementary Section S3). We therefore collected scattering patterns at different incident angles and measured the variation of the off-specular scattering intensity at a fixed (q_x, q_y) spot, as shown in Fig. 3a,b. The dependence of scattering intensity on incident angle demonstrates the effectiveness of data collection under the grazing-incidence condition, where the scattering intensity is orders of magnitude higher than that collected at higher incident angles.

To simulate the intensity curve in Fig. 3b, we developed a form-factor-independent model by combining the dynamical theory of reflectivity and the interference function of reflected waves. Because the goal of CDI is to obtain a reliable reconstruction without (or with minimal) a priori knowledge of the sample, our model does not account for any in-plane structural details of the sample. Instead, we directly model the interference of the ESWs at a certain point in the far field by simply assuming two scattered waves: one from gold and another from silicon. The interference of these two waves is calculated using Parratt formalism[16], with the path difference for the two waves propagating to the same spot on the detector. (Details of the model are provided in Supplementary Section S3.) The simulated result is shown in Fig. 3b as a solid line, and matches the data extremely well. With all the parameters obtained by data fitting, the angle-dependent

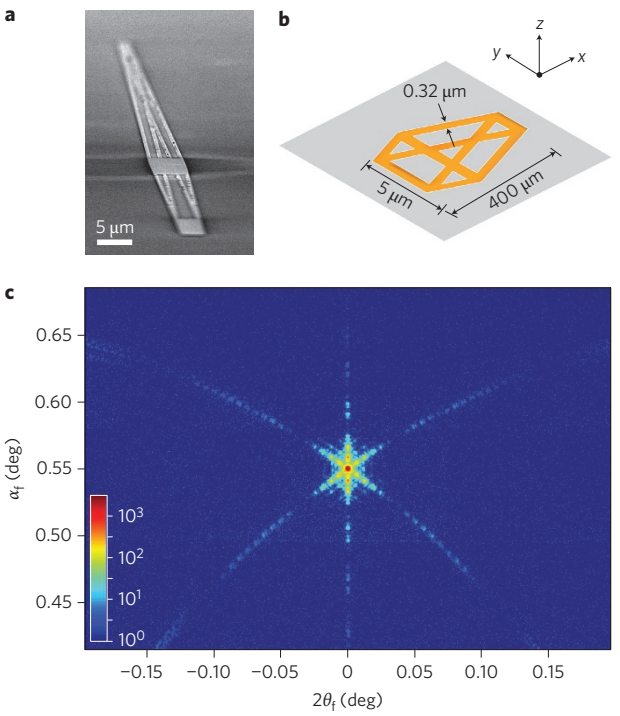


Figure 2 | SEM image of the sample and corresponding CSSI data. **a**, Angle-view SEM image of the sample with the Argonne National Laboratory logo pattern. **b**, Schematic of the pattern with major dimensions labelled. The thickness of the gold structure is 36 nm. **c**, Scattering pattern from the logo sample (logarithmic scale; units = counts), which is composed of stitched patterns collected at nine different detector positions.

© 2012 Macmillan Publishers Limited. All rights reserved.

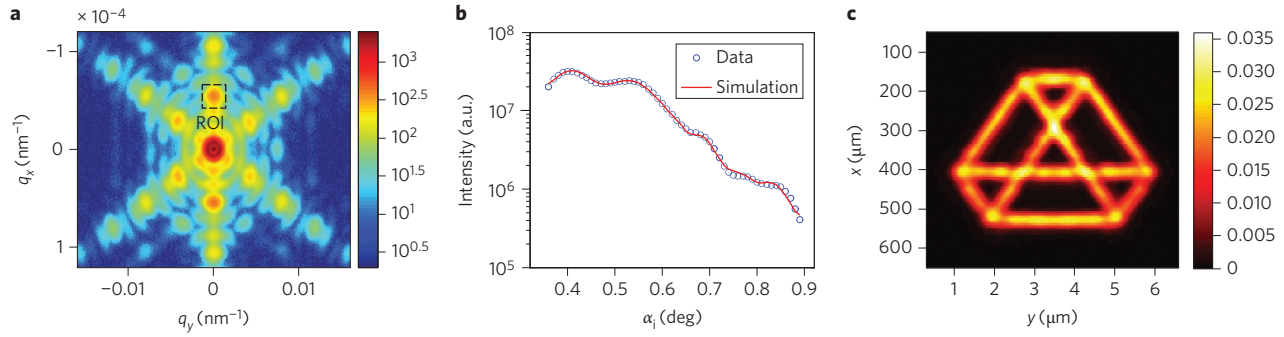


Figure 3 | Two-dimensional reconstruction of grazing-incidence CSSI data. a, Low-q region of the logo scattering pattern shown in Fig. 2c. ROI, region of interest. **b**, Experimental data (open circles) and simulation (solid line) of off-specular scattering intensity as a function of incident angle. Data were collected from a region of interest at a fixed (q_x, q_y) position with strong scattering, as shown in **a**. **c**, Reconstructed planar ESW amplitude of the logo structure. Note that pixel resolutions along the horizontal and vertical directions are different.

scattering contribution from the sample structure along the z-direction was calculated and used to normalize the two-dimensional scattering pattern for the constant-thickness structure. After q_x axis adjustment and q_z intensity normalization, reconstruction of the ESW amplitude was alternately performed using HIO and ER algorithms. The support was first estimated based on the autocorrelation function of the scattering pattern, and then gradually improved after a few iterations. Figure 3c shows the final result for the logo pattern with the aspect ratio adjusted to reflect the beam projection at the grazing-incidence angle.

As found in an abundance of traditional reflectivity experiments, the additional q_z dependence of the coherent scattering patterns during the angular scan strongly suggests that three-dimensional structural information is contained in the data set. To decipher the angular dependence, we collected a series of scattering data from the pattern at different incident angles ranging from 0.56° to 0.9°. The scattering patterns from lower incident angles were not counted for three-dimensional reconstruction because the Born approximation alone may not be sufficient to interpret the wave fields and scattering below the critical angles. The three-dimensional reciprocal-space data, shown in Fig. 4a, was constructed by filling the voxels in (q_x, q_y, q_z) space with the corresponding speckle intensity from each two-dimensional scattering pattern. Here, the three-dimensional reciprocal pattern is shown by an iso-surface rendering effect—low-intensity voxels do not show up in the plot, which leads to a seemingly smaller q range than in the actual data collected and used for reconstruction. The reconstructed image in two dimensions could be used as support for the x–y plane in the three-dimensional reconstruction. For the z-direction, the support dimension was estimated by a Fourier transform of the specular reflection intensity curve (that is, I_{specular} vs q_z) after multiplying by q_z^4, analogous to obtaining a thickness value from reflectivity data using the Fourier method[24]. The reconstructed three-dimensional image is shown in Fig. 4b,c (different viewing angles).

The magnitude of the ESW amplitude at each z-location (between the two arrows in Fig. 4c) was extracted and plotted in Fig. 4d. The error bar of each datum point represents the standard deviation of intensities from 300 pixel locations in the x–y plane. The intensity profile shows a peak with an extended plateau, which was fit by a flat-top Gaussian function, yielding a full-width at half-maximum (FWHM) of 36.9 nm, which is consistent with the thickness of the gold layer. The flat-top Gaussian profile is a model-independent fit to the ESW amplitude in the z-direction reconstructed from the limited q range involved in the inverse Fourier transform. This is analogous to reconstructing a square wave with truncated reciprocal data in the Fourier space. The slope on each side of the flat top represents the resolution of the height reconstruction, which should be approximately symmetrical. Even though the pattern studied here has a simple structure along the z-direction (that is, uniform thickness), there is no fundamental limit for applying the CSSI technique to reconstruct more complex three-dimensional architectures. However, scattering intensity redistribution and normalization will involve iterative application of the distorted-wave Born approximation[17,25] when the layered structure exhibits more complex scattering phenomena, such as wave-guide effects[26], near total external reflection.

At grazing incidence, spatial resolutions in the x-, y- and z-directions are highly anisotropic. Specifically, for a Coolsnap HQ^2 detector placed 0.7 m away from the sample, the x and y resolutions at an incident angle of 0.6° and with nine detector frame tiles are 2.6 μm and 22 nm, respectively, and a scan of incident angle to 0.9° gives a z resolution of 2.7 nm. As determined by the detectable q ranges, the real-space resolutions are limited by the coherent flux of the incident beam and the efficiency of the detector. Hypothetically, if the coherent flux is 1×10^{12} photons/s (10^3 times higher than the current case), the maximum q where the scattering intensity can be detected will be 10 times larger, leading to approximately 10 times higher real-space resolutions in all three directions. In addition, detectors with higher quantum efficiency and dynamic range and a lower noise level can improve the resolution. Even with these improvements, the resolution in the x-direction can still be one order of magnitude lower than that of the y-direction. One potential remedy is to take scattering images at several azimuthal angles around the sample normal direction to provide access to a wider in-plane q range for higher resolution. For structures composed of lighter elements, such as carbon, aluminium and iron, the detectable q range will be smaller than that of a strong scatterer like gold.

To conclude, we have developed a novel, optics-free, hard X-ray imaging technique by synergistically combining coherent surface scattering and grazing incidence near the sample total external reflection. By visualizing non-periodic gold nanostructures on a silicon substrate, we have demonstrated that CSSI is capable of reconstructing not only planar images of surface structures, but also three-dimensional architectures with nanometre resolution along the surface normal. Moreover, CSSI can be readily adapted to other coherent imaging techniques, such as resonant CDI, Fresnel CDI, holography, ptychography and ankylography, to study three-dimensional nanoscaled structures. Thanks to enhanced scattering near total external reflection, even with a hard X-ray coherent flux of $\sim 1 \times 10^9$ photons/s, our three-dimensional reconstruction of the images is practical. It should pave the way for nanostructure visualization in biology, materials science and condensed matter physics. With higher resolution it will find applications in time-resolved studies of biological membranes and supramolecules in an aqueous environment, thin film and quantum dot growth at

© 2012 Macmillan Publishers Limited. All rights reserved.

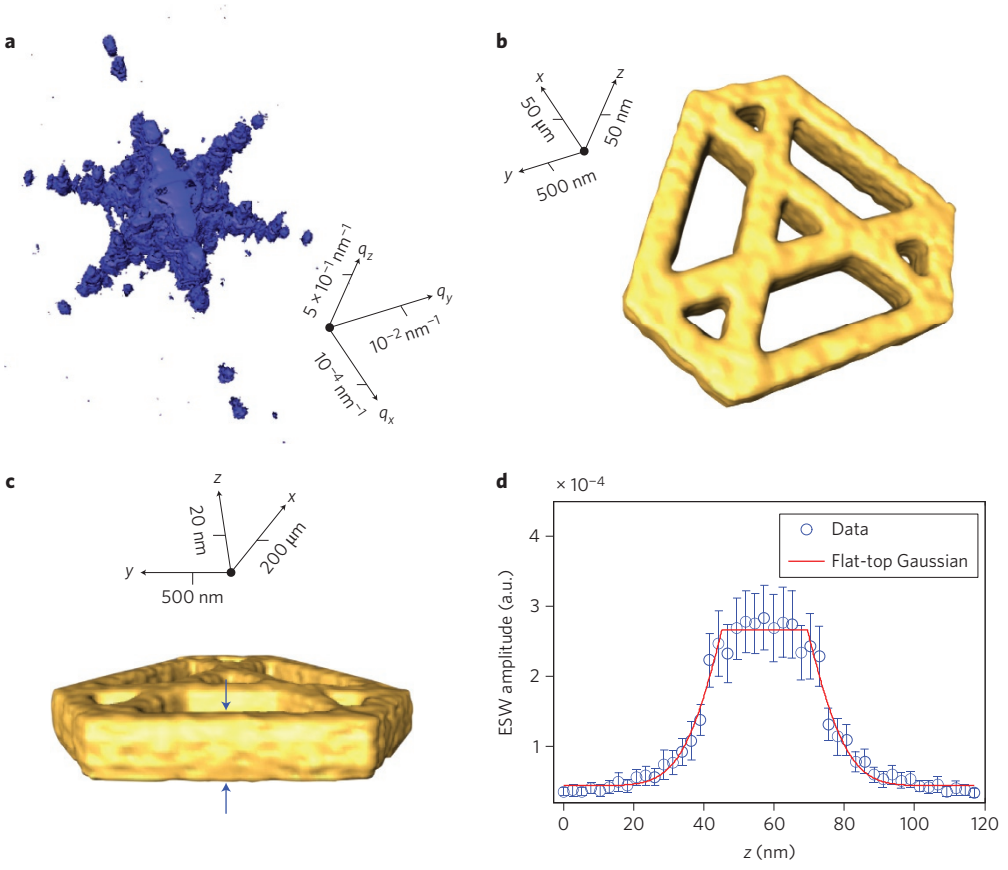


Figure 4 | Three-dimensional reconstruction of grazing-incidence CSSI data. a, Iso-surface rendering of the three-dimensional reciprocal data of the logo pattern, constructed from a series of two-dimensional scattering patterns obtained with different incident angles. **b**,**c**, Reconstructed three-dimensional ESW amplitude of the logo pattern from two different viewing angles. **d**, ESW amplitude as a function of different z (normal to the surface). The curve was fit using a flat-top Gaussian function with a FWHM of 36.9 nm which agrees well with the thickness of the gold layer. The error bar of each datum point represents the standard deviation.

surfaces and interfaces, and assembly of planar polymer nanocomposites[27]. In true reflection geometry, it can be a perfect means with which to study the mesophase liquid/solid interaction on nanopatterned grooved surfaces[28,29] and to perform non-destructive structural analysis of emerging three-dimensional nanocircuits integrated with thick wafers[30].

Methods

The samples studied in the present work were prepared in the cleanroom at the Center for Nanoscale Materials, Argonne National Laboratory. Nano- and microstructures of the gold top layer and a titanium adhesion layer were fabricated on flat silicon substrates using electron-beam lithography and sputtering techniques. The double-stripe pattern had a nominal line width of 850 nm, separation of 2 μm and length of 2 mm. The overall length of the logo pattern was 391 μm (x-direction), while the maximum width of the entire pattern was 5.1 μm (y-direction) and the width of the thinnest line in the pattern was 350 nm. For both samples, the thickness of the gold layer was ~36 nm and the titanium adhesion layer between the gold layer and the silicon substrate was ~7 nm thick.

The grazing-incidence CSSI experiments were performed at the 8-ID-E beamline of the Advanced Photon Source, Argonne National Laboratory. The monochromatic incident beam (photon energy $E = 7.35$ keV with $\Delta E/E = 2 \times 10^{-4}$) along the x-direction (see Fig. 1) was collimated by three sets of slit pairs to obtain an approximate Gaussian intensity distribution in the y–z plane with a FWHM of ~15 μm in both the horizontal and vertical directions. The transverse coherence length along the horizontal direction was ~7 μm and was fully coherent across the entire vertical direction over the 15 μm beam. The longitudinal coherence length in our experiment was ~1 μm, sufficient for the grazing-incidence small-angle scattering geometry. A detailed discussion on coherence is included in Supplementary Section S5. The flux was typically 1×10^9 photons/s for all experiments. The incident angles ranged from 0.35 to 0.9°, around the critical angle of gold ($\alpha_c = 0.61°$ at $\lambda = 0.1687$ nm). A Coolsnap HQ^2 detector (1,040 × 1,392 pixels, 6.45 × 6.45 μm^2 per pixel and 12-bit readout) was placed ~0.7 m away from the sample for collecting scattering patterns. A ×5 lens was used, resulting in an effective pixel dimension of 1.29 × 1.29 μm^2 on the detector. The scintillator crystal was a 5-μm-thick film of yttrium aluminium garnet doped with cerium (YAG:Ce). The overall detection efficiency of the detector system was estimated to be 0.13% (~750 photons/count). The scattering pattern from the double-stripe structure was averaged over 40 exposures with 5 s exposure time each, and the scattering pattern from the logo structure was averaged over five exposures with 45 s exposure time each.

Our CSSI set-up had no optics between the slits and the detector, allowing our imaging technique to be easily adopted by any hard or soft X-ray scattering beamline where coherence of the source is preserved. In the grazing-incidence geometry, the sample functions as the beam stop, because the strong incident beam is sufficiently attenuated by the substrate. Therefore, extremely low-q ($q_x \approx 7.6 \times 10^{-6}$ nm^{-1}, $q_y \approx 7.6 \times 10^{-4}$ nm^{-1}) scattering intensity is available in CSSI providing the 'support' or maximum feature size during both two- and three-dimensional reconstructions.

For phase retrieval, each reconstruction contained 100 iterations, which alternated between nine applications of the HIO algorithm and one application of the ER algorithm (except for the last iteration). During iteration, those pixels with an intensity lower than a certain threshold, including the void voxels in a three-dimensional reciprocal pattern, were masked out and allowed to float. Note that the data for three-dimensional reconstruction were multiplied by q_z^2 before performing reconstruction as a calibration of the algebraic decay of the diffuse scattering intensity with q_z (ref. 17). The positivity constraint was also applied for the three-dimensional reconstruction. The final results for both two- and three-dimensional cases were averaged over 100 total reconstructions, each starting with a random phase and random value of the feedback parameter β in the HIO algorithm ($\beta \approx 0.6$–0.9), and then normalized with the intensity profile of the incident beam. In our experiment, because the beam size was larger than the transverse coherence length in the y-direction, the 'specular' reflection spot contained incoherent and coherent scattering intensity beyond the instrument resolution. The specular reflection region was therefore also masked out using a numerical beam stop with a maximum diameter of 25 pixels. Even so, the scattering pattern provided sufficient low-q information ($q_x \approx 1 \times 10^{-5}$ nm^{-1}, $q_y = 8.8 \times 10^{-4}$ nm^{-1}).

© 2012 Macmillan Publishers Limited. All rights reserved.

For two-dimensional imaging, two competing factors need to be considered when selecting the incident angle for a particular sample structure: detectable q_x range and off-specular scattering intensity. In grazing-incidence geometry, a smaller incident angle yields a smaller range of q_x, and therefore lower real-space resolution in the x-direction. Thus, a relatively large incident angle is favourable. Plots of the detectable q range at different incident angles are shown in Supplementary Section S1. On the other hand, because the scattering intensity decreases with q_z, with a relatively low incident coherent photon flux as in this study, a smaller incident angle is desirable to achieve a high signal-to-noise ratio in a reasonable amount of data collection time. Taking the logo structure as an example, the optimized incident angle for the logo structure lies around 0.55° in our experiment, where the q range is large enough to give the necessary x resolution and the off-specular scattering intensity reaches a local maximum due to a constructive interference between the surface scattering signals from the sample.

Received 6 May 2012; accepted 22 June 2012; published online 12 August 2012

References

1. Miao, J. W., Charalambous, P., Kirz, J. & Sayre, D. Extending the methodology of X-ray crystallography to allow imaging of micrometre-sized non-crystalline specimens. *Nature* **400,** 342–344 (1999).
2. Eisebitt, S. *et al.* Lensless imaging of magnetic nanostructures by X-ray spectro-holography. *Nature* **432,** 885–888 (2004).
3. Shapiro, D. *et al.* Biological imaging by soft X-ray diffraction microscopy. *Proc. Natl Acad. Sci. USA* **102,** 15343–15346 (2005).
4. Pfeifer, M. A., Williams, G. J., Vartanyants, I. A., Harder, R. & Robinson, I. K. Three-dimensional mapping of a deformation field inside a nanocrystal. *Nature* **442,** 63–66 (2006).
5. Chapman, H. N. *et al.* Femtosecond time-delay X-ray holography. *Nature* **448,** 676–679 (2007).
6. Abbey, B. *et al.* Keyhole coherent diffractive imaging. *Nature Phys.* **4,** 394–398 (2008).
7. Thibault, P. *et al.* High-resolution scanning X-ray diffraction microscopy. *Science* **321,** 379–382 (2008).
8. Raines, K. S. *et al.* Three-dimensional structure determination from a single view. *Nature* **463,** 214–217 (2010).
9. Giewekemeyer, K. *et al.* Quantitative biological imaging by ptychographic X-ray diffraction microscopy. *Proc. Natl Acad. Sci. USA* **107,** 529–534 (2010).
10. Abbey, B. *et al.* Lensless imaging using broadband X-ray sources. *Nature Photon.* **5,** 420–424 (2011).
11. Robinson, I. & Harder, R. Coherent X-ray diffraction imaging of strain at the nanoscale, *Nature Mater.* **8,** 291–298 (2009).
12. Chapman, H. N. & Nugent, K. A. Coherent lensless X-ray imaging. *Nature Photon.* **4,** 833–839 (2010).
13. Yefanov, O. M. *et al.* Coherent diffraction tomography of nanoislands from grazing-incidence small-angle X-ray scattering. *Appl. Phys. Lett.* **94,** 123104 (2009).
14. Marathe, S. *et al.* Coherent diffraction surface imaging in reflection geometry. *Opt. Express* **18,** 7253–7262 (2010).
15. Roy, S. *et al.* Lensless X-ray imaging in reflection geometry. *Nature Photon.* **5,** 243–245 (2011).
16. Parratt, L. G. Surface studies of solids by total reflection of X-rays. *Phys. Rev.* **95,** 359–369 (1954).
17. Sinha, S. K., Sirota, E. B., Garoff, S. & Stanley, H. B. X-ray and neutron-scattering from rough surfaces. *Phys. Rev. B* **38,** 2297–2311 (1988).
18. Schlomka, J. P. *et al.* X-ray-diffraction from Si/Ge layers—diffuse-scattering in the region of total external reflection. *Phys. Rev. B* **51,** 2311–2321 (1995).
19. Tolan, M., Press, W., Brinkop, F. & Kotthaus, J. P. X-ray-diffraction from laterally structured surfaces—total external reflection. *Phys. Rev. B* **51,** 2239–2251 (1995).
20. Jiang, Z., Lee, D. R., Narayanan, S., Wang, J. & Sinha, S. K. Waveguide-enhanced grazing-incidence small-angle X-ray scattering of buried nanostructures in thin films. *Phys. Rev. B* **84,** 075440 (2011).
21. Dutta, P. & Sinha S. K. Analytic form for the static structure factor for a finite two-dimensional harmonic lattic. *Phys. Rev. Lett.* **47,** 50–53 (1981).
22. Fienup, J. R. Reconstruction of an object from modulus of its Fourier transform. *Opt. Lett.* **3,** 27–29 (1978).
23. Gerchberg, R. W. & Saxton, W. O. A practical algorithm for the determination of phase from image and diffraction plane pictures. *Optik* **35,** 237–246 (1972).
24. Seeck, O. H. *et al.* Analysis of X-ray reflectivity data from low-contrast polymer bilayer systems using a Fourier method. *Appl. Phys. Lett.* **76,** 2713–2715 (2000).
25. Lazzari, R. IsGISAXS: a program for grazing-incidence small-angle X-ray scattering analysis of supported islands. *J. Appl. Crystallogr.* **35,** 406–421 (2002).
26. Wang, J., Bedzyk, M. J. & Caffrey, M. Resonance-enhanced X-rays in thin-films—a structure probe for membranes and surface-layers. *Science* **258,** 775–778 (1992).
27. Lin, Y. *et al.* Self-directed self-assembly of nanoparticle/copolymer mixtures. *Nature* **434,** 55–59 (2005).
28. Hofmann, T. *et al.* Wetting of nanopatterned grooved surfaces. *Phys. Rev. Lett.* **104,** 106102 (2010).
29. Reyssat, M., Yeomans, J. M. & Quere, D. Impalement of fakir drops. *Europhys. Lett.* **81,** 026006 (2008).
30. Xie, Y., Cong, J. & Sapatnekar, S. *Three-Dimensional Integrated Circuit Design: EDA, Design and Microarchitectures* (Springer, 2009).

Acknowledgements

The authors thank X. Huang for constructive suggestions, and M. Guizar, S. Sinha, C. Jacobson, A. Sandy and S. Narayanan for valuable discussions. The authors also thank A. Khounsary for providing the ultraflat silicon substrate, and B. Liu, A. Yan, V. Dravid, J. Qian, L. Assoufid, R. Divan, D. Rosenmann, C. Liu and M. Wieczorek for their assistance with sample fabrication and characterization. R. Bradford, T. Lutes and M. Rivers are thanked for their cooperation with detector usage. Use of the Advanced Photon Source and the Center for Nanoscale Materials were supported by the US Department of Energy Office of Science (contract no. DE-AC02-06CH11357).

Author contributions

J.W. conceived the project. T.S., Z.J. and J.W. designed the experiment and participated in data analysis. T.S., Z.J. and J.S. conducted the experiment. Samples were fabricated by T.S. and L.O. All authors discussed the results and contributed to writing the manuscript.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permission information is available online at http://www.nature.com/reprints. Correspondence and requests for materials should be addressed to J.W. and T.S.

Competing financial interests

The authors declare no competing financial interests.

© 2012 Macmillan Publishers Limited. All rights reserved.